Project Demo Planning

Date	29/06/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Present project title 'Rising Waters', team of 5 members, and the flood prediction problem. Explain why manual meteorological analysis is too slow, the impact of floods in India, and how ML addresses the gap. Briefly state XGBoost achieves 96.55% accuracy.	2 mins	Gowthamsai Setti (Team Lead, Sprint-1 Lead)
2	Dataset & EDA Walkthrough	Open flood_eda.ipynb. Show flood dataset.xlsx — 10 features (Temp, Humidity, Cloud Cover, ANNUAL, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec, avgjune, sub) and target variable (flood). Walk through univariate analysis (distribution plots, box plots), multivariate analysis (correlation heatmap, violin plots) and descriptive statistics (head(), shape, info(), describe()).	3 mins	Karri Dhinesh (Sprint-1: EDA) Rishitha Padaga (Sprint-1: Univariate)
3	Data Preprocessing & Model Building	Open flood_model.ipynb. Demonstrate preprocessing pipeline: missing value handling (isnull().sum(), fillna()), IQR-based outlier capping, Label Encoding, 80/20 stratified train-test split, and StandardScaler normalization. Show training of all 4 classifiers (Decision Tree, Random Forest, KNN, XGBoost). Present confusion matrix, F1-score and ROC-AUC comparison table. Explain why XGBoost with scale_pos_weight was selected.	4 mins	Sai Siddardha Bevara (Sprint-2 Lead & Sprint-3: Model Building) Gowthamsai Setti (Sprint-2 & Sprint-3)
4	Live Web App Demonstration	Open https://rising-waters-flood-prediction.onrender.com. Show home.html landing page. Navigate to /Predict (index.html) and enter sample flood data: Temp=31, Humidity=79, Cloud Cover=44, ANNUAL=4000, Jan-Feb=90, Mar-May=800, Jun-Sep=3000, Oct-Dec=700, avgjune=350, sub=900. Submit and show chance.html result with probability score, animated progress bar and advisory recommendations. Repeat with low-risk values to show no_chance.html.	4 mins	Rishitha Padaga (Sprint-4: HTML Templates & Testing)
5	Code, GitHub & Deployment Walkthrough	Show project file structure: app.py (4 Flask routes), Procfile (web: gunicorn app:app), requirements.txt (10 dependencies), floods.save, transform.save, static/css/main.css (dark-themed rain animation), static/js/main.js. Open GitHub repo (github.com/Gowthamsai1234/rising-waters-flood-prediction) and show commit history. Explain Render.com auto-deploy pipeline triggered on git push to main branch.	3 mins	Ravi Kumar Yadla (Sprint-4: Deployment) Gowthamsai Setti (Sprint-4: Flask Integration)

S.No	Demo Section	Description	Duration (mins)	Responsible Member
6	Performance Testing & Q&A; Session	Show flood_test.js k6 script — 3 stages (ramp-up 0→10 VUs, peak 20 VUs, ramp-down), testing GET /, GET /Predict and POST /Predict. Present k6 results: 95th percentile response time < 2s, error rate < 1%. Open floor for evaluator Q&A; — each member answers questions in their sprint domain.	4 mins	All 5 Members (Ravi Kumar Yadla: Performance Testing Lead)

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Gowthamsai Setti opens the demo. Introduces all 5 team members, project title, and the flood prediction problem in India. States the goal: XGBoost-based real-time flood risk classification from 10 meteorological inputs.
2	Solution Overview	Brief walkthrough of the complete ML pipeline: flood dataset.xlsx → EDA (flood_eda.ipynb) → Preprocessing + Model Training (flood_model.ipynb) → Flask app (app.py) → Render.com deployment. Highlight 96.55% XGBoost accuracy and live public URL.
3	Live Feature Demonstration	Rishitha opens live app on Render.com. Karri presents EDA notebook visualizations. Sai Siddardha presents model comparison and XGBoost results. Ravi shows GitHub repo structure and Render.com deployment logs. Gowthamsai demonstrates Flask routes and model integration in app.py.
4	Q&A; Session	All 5 members respond to evaluator questions by domain: EDA/data (Karri Dhinesh, Rishitha Padaga), preprocessing/model (Sai Siddardha Bevara), deployment/CI-CD (Ravi Kumar Yadla), Flask integration/overall (Gowthamsai Setti). Performance test results (k6) presented if asked.